



Biology, Geology and Environmental Sciences · 1st year of Bachillerato · Inside the Earth · Class 1 of 5 · With Nil

# The Earth is too heavy



Class video: <https://youtu.be/s2rJ18BUAQ0>

The code leads to the video.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Before you start

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1. Draw the layers of the Earth you learned. How do you know they are like that?
2. **Your prediction.** The whole Earth, on average: is it denser than a stone from the street, equally dense, or less dense?

## While you watch the video

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1. Density is mass divided by volume. Why, when you lower the hanging stone into the water, does the scale show its volume?
2. **The control.** Half a kilo of steel pushes aside 64 g of water. What density does that give? What does it tell you about the setup?
3. If the Earth is twice as dense as the rocks we walk on, where is the weight?

## Your measurements

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| Sample          | Dry mass (g) | Reading when immersed (g) | Density (g/cm <sup>3</sup> ) | Interval |
|-----------------|--------------|---------------------------|------------------------------|----------|
| Steel (control) |              |                           |                              |          |
|                 |              |                           |                              |          |
|                 |              |                           |                              |          |
|                 |              |                           |                              |          |

## Practice

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1. A 180 g stone pushes aside 62 g of water. What is its density? Between which values is it if the scale is off by a gram? What rock could it be?
2. What fraction of the Earth's radius (6,371 km) is the deepest mine, about 4 km deep?
3. Explain with a number why the Earth cannot be hollow.

## Keep thinking

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Can you be sure of something nobody will ever see?